

Primordial Gravitational Waves as a Viable Explanation of The observation of $z \approx 0.5$ Gradient Reversal in DESI

-A conjecture: Cosmic-QCD mirroring

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Abstract:

The observed gradient reversal in the dark energy density distribution across the $z \approx 0.5$ redshift region of the paper "Extended Dark Energy analysis using DESI DR2 BAO measurements" could be explained with primordial gravitational wave.

The observed gradient reversal in the dark energy density distribution across the $z \approx 0.5$ redshift region, as presented in Figure 2 of the DESI paper "Extended Dark Energy analysis using DESI DR2 BAO measurements" DOI:10.48550/arXiv.2503.14743[1], suggests a global-scale mechanism rather than localized effects. Since the gradient manifests as a universal feature across cosmic scales, the influence of local strong gravitational fields on dark energy density distribution can be excluded. Among potential global-scale factors, primordial gravitational waves generated during the universe's inception merit particular consideration. The alternating compression and rarefaction of spacetime induced by gravitational wave propagation (corresponding to density enhancement and reduction respectively) could combine with cosmic expansion to produce the observed large-scale dark energy density variation.

The $z \approx 0.5$ redshift region might represent a phase-critical boundary where the primordial gravitational waves transition between compressive and tensile dominance. This interpretation aligns with the observed reversal in dark energy density trends: ascending density distribution at $z < 0.5$ transitions to descending distribution at $z > 0.5$.

The asymmetric profile in Figure 2 of DESI paper further implies a possible non-reciprocal response of spacetime to gravitational wave strains, analogous to differing elastic moduli under compression versus tension in material systems and mirroring QCD quark confinement/asymptotic freedom? (Cosmic-QCD mirroring effect).

The Cosmic-QCD mirroring effect proposed herein postulates that spacetime's nonlinear elastic response to primordial gravitational waves (manifested as redshift-dependent compression tension asymmetry) exhibits a phase transition behavior analogous to QCD's confinement-asymptotic freedom dichotomy.

Notably, the steeper gradient observed at $z < 0.5$ suggests enhanced sensitivity to compressive perturbations when combined with the baseline density reduction from cosmic expansion.

This analysis proposes three hypotheses based on The elastic spacetime paradigm of GR

1) Primordial gravitational wave-induced spacetime Distortion exhibits intrinsic asymmetry between compressive and tensile modes, that could be intrinsic feature of the universe spacetime.

2) The interaction between wave-induced Distortion fields and cosmological expansion produces redshift-dependent modulation of dark energy density distribution.

3) Space's elastic moduli under compression versus tension might be non-reciprocal.

Reference:

[1] Extended Dark Energy analysis using DESI DR2 BAO measurements

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